DeepFake Analysis Report

Explainable Deepfake Detection — Xception-LSTM with LIME · Research prototype

Analysis ID	dcbe720a-5cde-4078-9944-6920f855a1f1
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1. Executive summary

REAL	raw fake score 0.325892	decision threshold 0.6 (dev-v1)	model version v2

fake_score 0.325892 vs decision threshold 0.6 (dev-v1)

This score is an uncalibrated sigmoid output — it is not a probability and not forensic proof.

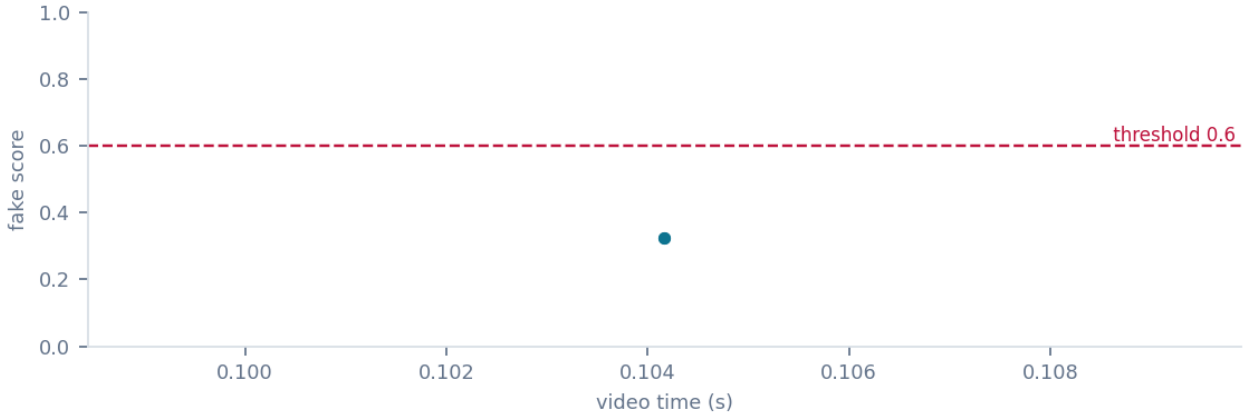
2. Media information

Filename	test_video.mp4
Media type	video
File size	164.2 KB
Resolution	1920×1080
Duration	0.25 s
FPS	24.0
SHA-256	b3d47c8919ae58ae2e0ca219b04a59f1dadfe72d37519087c03dd9d371124a49
Frames extracted / analyzed	5 / 5

3. Prediction analysis

Raw model output (fake score)	0.325892
Decision method	Sliding-window ensemble (full 32-frame score + 10-frame windows, averaged)
Decision threshold	0.6 (dev-v1)
Score calibrated?	No — sigmoid output, uncalibrated
Model version	v2
Window score variation	min 0.3259 · max 0.3259 · stdev 0.0000

4. Video prediction timeline



Window scores are plotted against the source video timeline using fps and frame indices captured during face extraction. The dashed line marks the active decision threshold.

5. Visual evidence



Processed face (elliptical mask applied)



LIME explanation heatmap

LIME highlights regions of a single replicated frame that most influenced the model's response. It does NOT prove those pixels were manipulated and does not explain the temporal (LSTM) pathway.

6. Technical details

Model	Xception-LSTM · version v2
Checkpoint SHA-256	9ee14530166b842ce0aa6ac74500ac138993f5fbe82a927df7ddee0ffc261228
Sequence length	32 frames @ 299×299 px
Preprocessing	OpenCV DNN face detection (confidence 0.5) → 20% margin crop → 299×299 resize → elliptical background mask → [0, 1] normalization
Prediction method	Sliding-window ensemble (full 32-frame score + 10-frame windows, averaged)
Processing duration	191.9 s

7. Interpretation

The model scored this media below the configured decision threshold, which maps to its REAL class. The score falls short of the threshold by 0.274. Scoring method: sliding_window_ensemble. Because the model is uncalibrated and was evaluated as weak on held-out data (see evaluation/EVALUATION.md), this assessment must be treated as a research-grade indication and reviewed by a human before any consequential use. Quality warnings were raised during processing.

8. Warnings

• Only 5 face frames extracted (model expects 32); sequence was padded by repeating the last frame, which can weaken reliability.

9. Research limitations

1. Scores are sigmoid outputs from an uncalibrated model; they are not proven probabilities.

2. Models were trained only on the DeepFake Detection (DFD) subset of FaceForensics++; performance on other manipulation methods is unknown.

3. The train/val split has identity leakage, so measured metrics are optimistic.

4. Only the most confident face per frame is analyzed; multi-face videos are not fully covered.

5. LIME explains a replicated static frame; it does not attribute the temporal pathway.

6. Face-detection variability across OpenCV builds can change scores for the same video.